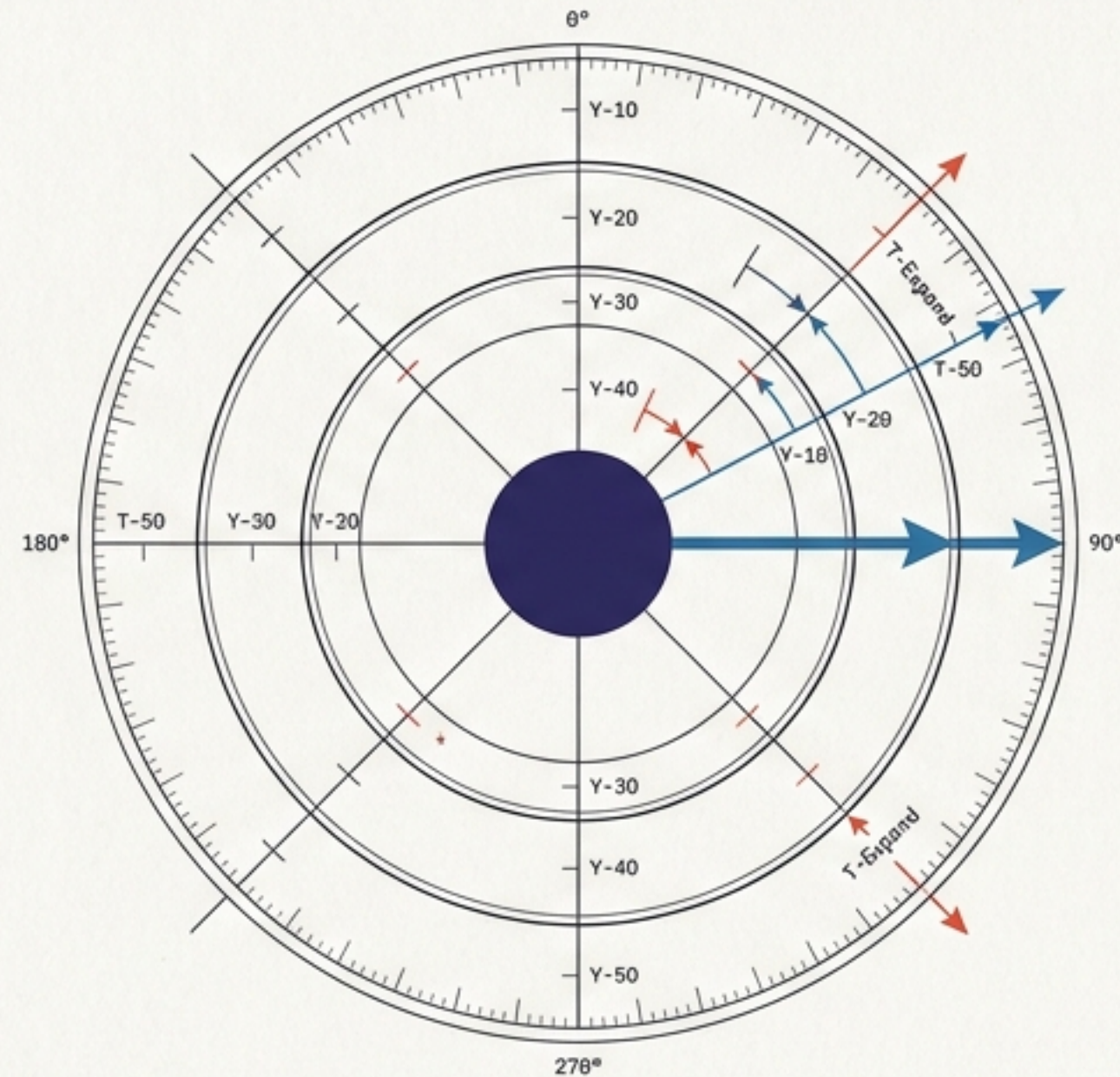


The Time Machine

Design Documentation: LinkFamiliesByDemographics.py



Bypassing the 75% Nameless Data Problem

Phase 1

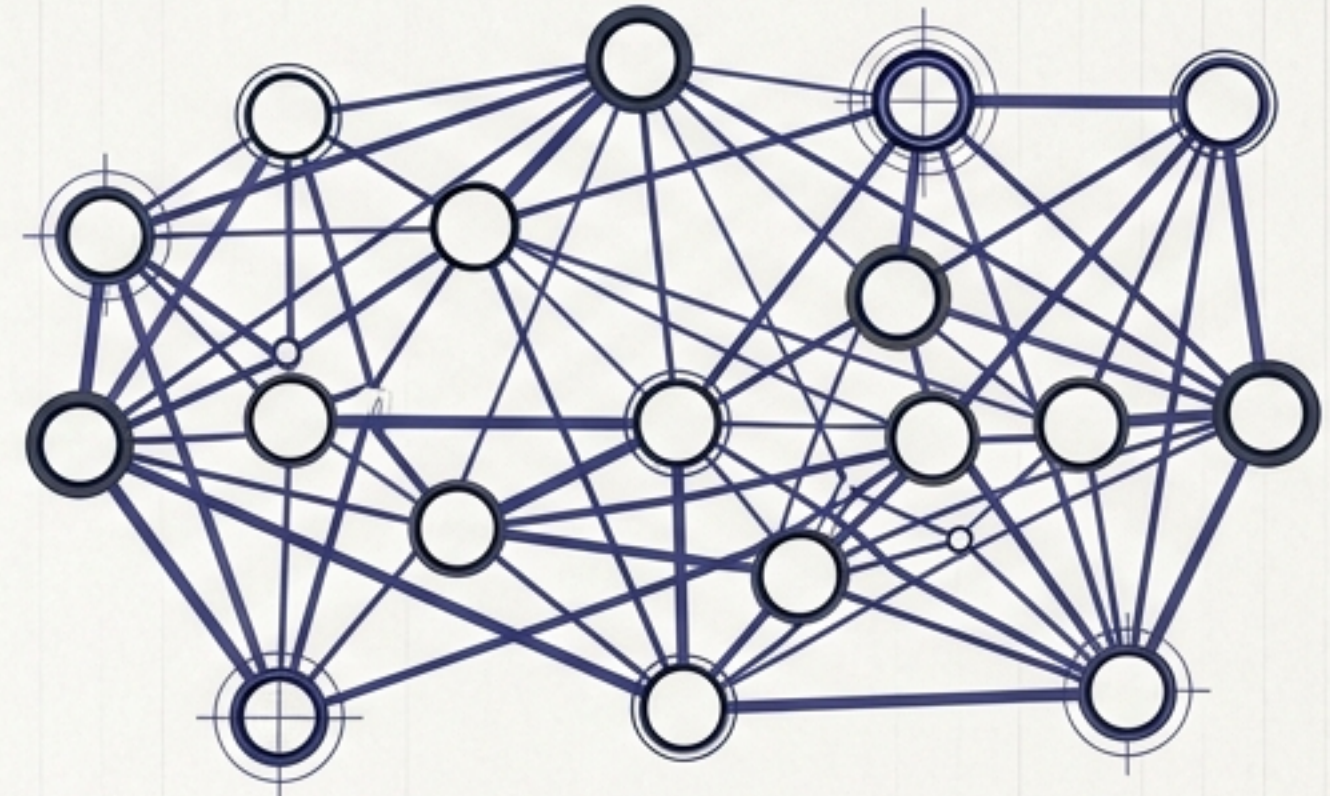


25% Coverage

Phase 1: Splink AI

Excels at resolving probabilistic 19th-century spelling errors (e.g., Jaro-Winkler distance), but is completely blind to blank records.

Phase 1a

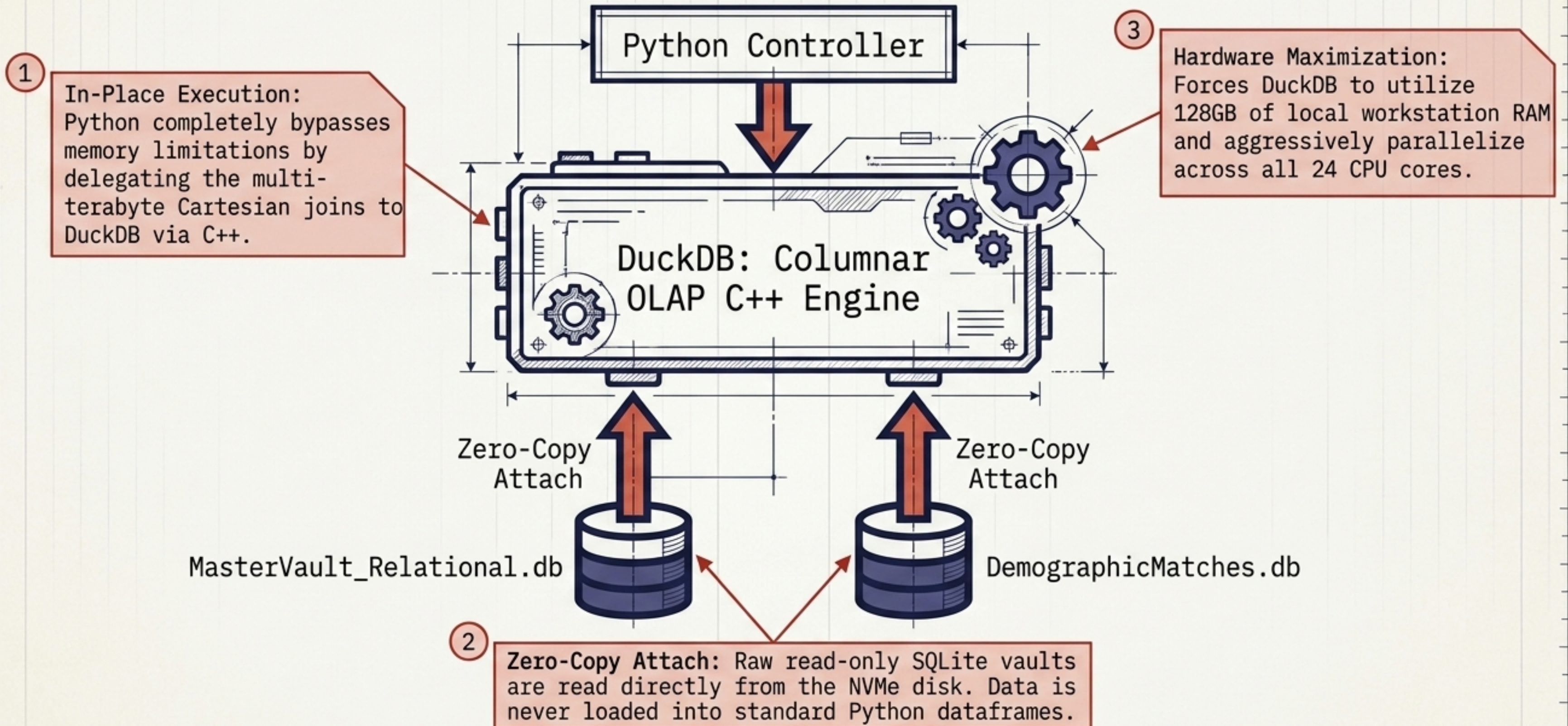


75% Coverage

Phase 1a: The Time Machine

Relies entirely on pure relational graph theory and household demographic vectors, allowing us to mathematically link millions of anonymous families across time without a single name.

The Engine: DuckDB OLAP Setup



Step 1: Extracting the 10-Variable Fingerprint

	Head of Household	Spouse
Sex	Male	Male
Birth Year	1840	1840
BPLD (Own Birthplace)	PA	PA
FBPL (Father's Birthplace)	NY	NY
MBPL (Mother's Birthplace)	MA	MA

1

The Problem:

The Problem: A single [Male, Born 1840, PA] returns thousands of false positives.

2

The Solution: By flattening the Head and Spouse demographics into a single row per family_id, the script creates an unbreakable, name-free cryptographic hash. The odds of two completely different households sharing the exact same 10-variable demographic combination are practically zero.

BPL (general)

BPLD (detailed)

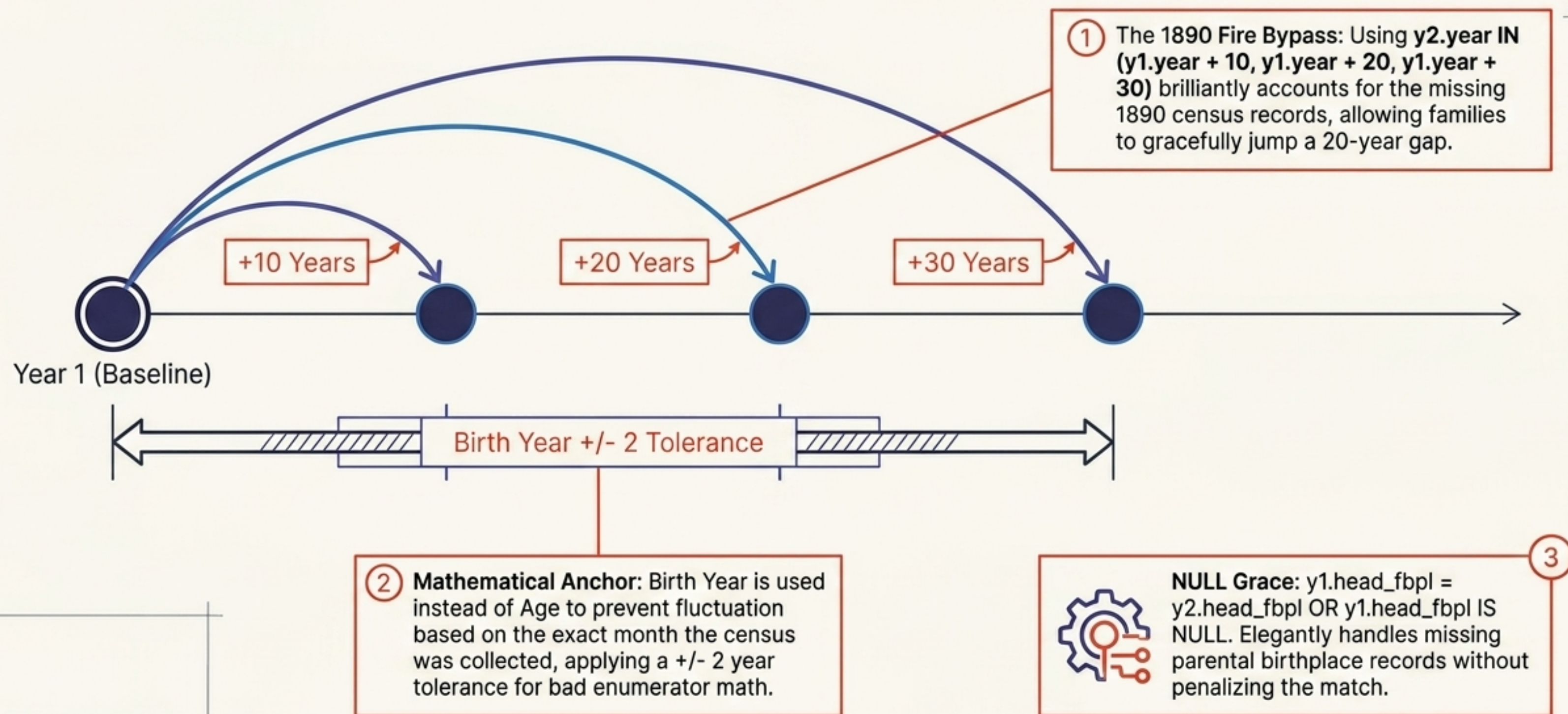
MBPL (general)

Flattened Household Vector
(hh_features TEMP table)

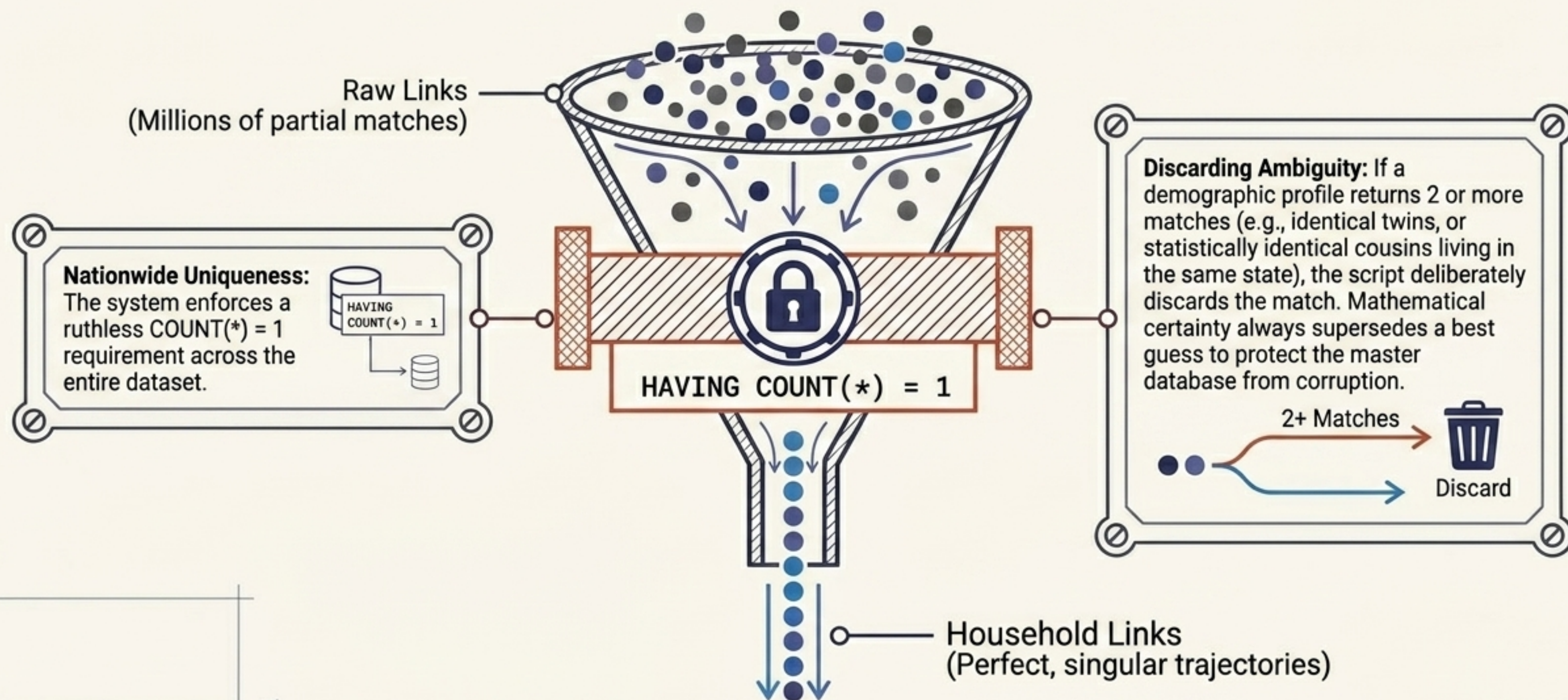
FBPL (general)

FBPLD (detailed)

Step 2: Cross-Decade Joining Logic



Step 3: Strict 1-to-1 Uniqueness Constraints



System Synthesis: End-to-End Execution



The Output:

The Time Machine successfully outputs **mathematically linked, multi-decade family vectors** without utilizing a single text string.

Next Steps:

These perfect matches are now structurally prepared to be processed by the Union-Find graph algorithm, dynamically grouping them into universal FAM_ buckets for final GEDCOM export.